



Data Article

Dataset for orange fruit detection from UAV in citrus orchards



Guillem Montalbán-Faet, Enrique Navarro-Modesto,
Andoni Salcedo-Navarro, Rafael Fayos-Jordan,
Pablo Benlloch-Caballero, Miguel García-Pineda*,
Jaume Segura-García*

Computer Science Department, ETSE-UV, Universitat de València, València, Spain

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ABSTRACT

Accurate fruit detection in citrus orchards is essential for yield estimation, precision harvesting, and automated orchard monitoring. Although UAV-based imaging has become a powerful tool in precision agriculture, publicly available datasets for orange fruit detection remain scarce, particularly those integrating multispectral data under real field conditions. This lack of open resources limits the development and benchmarking of robust deep-learning models for cross-spectral and illumination-invariant detection.

We present CampanetaOrangeFruit, a dataset acquired with a DJI Mavic 3 Multispectral UAV flying at 14 m above ground level over a commercial citrus orchard in Corbera, Valencia, Spain. The dataset comprises 550 synchronized captures (RGB + four multispectral bands: R, G, RE, NIR) for a total of 2750 images and 301,232 annotated orange instances. Each image includes YOLOv5-format annotations generated through a homography-based reprojection process, ensuring geometric consistency across spectral modalities.

CampanetaOrangeFruit uniquely provides pixel-aligned, cross-spectral UAV imagery with fine-grained fruit-level annotations, enabling research on fruit detection, yield estimation, and domain adaptation in real-world orchard environments. It represents a valuable benchmark for advancing

* Corresponding authors.

E-mail addresses: miguel.garcia-pineda@uv.es (M. García-Pineda), jaume.segura@uv.es (J. Segura-García).

Specifications Table

Subject	Computer Science
Specific subject area	Citrus Detection, Precision Agriculture, Remote Sensing, Photogrammetry
Type of data	Image files (raw JPG/TIFF), annotation text files, scripts and figures
Data collection	• Platform: DJI Mavic 3 Multispectral UAV • Altitude: 14 m Above Ground Level • Speed: 1.2 m/s • Overlap: 80% frontal, 70% lateral • Ground Sampling Distance: 0.65 cm/pixel
Data source location	Corbera, Valencia, Spain {39.176902, -0.353776; 39.176283, -0.351407}
Data accessibility	10.5281/zenodo.17866343 under CC-BY-4.0 licence
Related research article	The associated research article with this dataset is under preparation

1. Value of the Data

- The Campaneta-Orange-Fruit dataset provides a large-scale UAV-based collection of RGB and multispectral imagery of orange orchards acquired under real field conditions, offering a unique aerial perspective that enables detailed fruit-level analysis rarely available in open-access agricultural datasets.
- It comprises 2750 pixel-aligned images across five spectral bands (RGB, G, R, RE, NIR) and over 301,000 annotated oranges, representing one of the most extensive public resources for citrus fruit detection, yield estimation, and precision horticulture research. he repository has an approximate size of 21 GB.
- The multispectral configuration supports the study of cross-spectral modeling, illumination invariance, and canopy–fruit reflectance interactions, expanding the potential of deep-learning models to generalize under diverse spectral and lighting conditions.
- Instance-level YOLO annotations and accompanying metadata, flight logs, calibration parameters, and Python scripts for homography estimation and label reprojection ensure geometric consistency across modalities and promote full reproducibility in cross-spectral experiments.
- Potential downstream applications include automated fruit counting, yield mapping, and precision harvesting, advancing sustainable orchard management and UAV-based monitoring for intelligent, data-driven agriculture.
- The dataset supports practical yield estimation and production quantification for citrus growers by enabling automated fruit detection and counting at tree and orchard scale. Such UAV-based production assessment can assist farmers in harvest scheduling, labor and logistics planning, and market forecasting, contributing to data-driven decision-making and improved economic sustainability in commercial orchards.

2. Background

Precision agriculture increasingly leverages remote sensing to optimize orchard management, enabling non-destructive monitoring of fruit load, canopy health, and harvest timing in a cost-efficient and scalable way [1].

Recent advances in digital agriculture technologies—including 5G-IoT communication frameworks, intelligent sensor networks, and interactive visualization platforms—are further transforming precision farming by enabling real-time data acquisition, secure connectivity, and scalable analytics for decision support [2–4]. These innovations integrate UAV imagery, ground sensors, and cloud infrastructures into unified monitoring systems, enhancing automation, interoperability, and responsiveness across agricultural environments.

Unmanned aerial vehicles (UAVs) equipped with multispectral or RGB sensors have become key tools in orchard-scale monitoring because they deliver high spatial resolution and flexible deployment compared with satellite or ground-based methods [5,6].

In citrus orchards, UAV-based imagery has been successfully used for fruit detection, fruit counting, and yield estimation, providing accurate, non-invasive measures of production potential [7–11]. Datasets such as CitDet, YOLO-tiny, and YOLOv8-Scm have enabled model benchmarking for fruit counting, ripeness recognition, and even sunburn detection under field conditions [12–14]. Direct fruit-detection approaches from aerial imagery support pre-harvest yield prediction and facilitate decision-support systems in commercial orchards [9,11,12].

Multispectral imaging — capturing not only RGB but also red-edge and near-infrared bands — provides additional spectral information that helps discriminate fruits, canopy, and background, improving robustness to illumination changes and occlusions [15,16]. Integrating these narrow-band cues with vegetation indices or 3-D canopy structure enhances feature extraction and overall detection accuracy [13,17].

Recent deep-learning frameworks, including improved YOLOv7–v9 models, convolutional networks, and attention-based architectures, have demonstrated excellent performance for orchard imagery analysis, enabling the detection of individual fruits and estimation of yield at tree or orchard level [7,9,11,14,21]. These methods are further strengthened through multisensor fusion and IoT-based UAV networks, which allow real-time diagnosis and disease detection in citrus orchards [18,19].

Despite these advances, there remains a shortage of publicly available, high-resolution UAV datasets for citrus fruit detection that combine multispectral data, multi-temporal acquisitions, and instance-level annotations. Existing repositories, such as CampanetaWeed [20] or EscaYard [6], focus on weed or disease detection but rarely include fruit-specific labeling or pixel alignment. Other datasets, such as CitDet [12] or SootyMold [15], provide valuable RGB data but lack multi-band spectral coverage.

To fill this gap, the CampanetaOrangeFruit dataset provides pixel-aligned RGB + multispectral (R, G, RE, NIR) imagery of citrus orchards captured over UAV flight. The dataset includes instance-level annotations of individual oranges, enabling robust research on fruit detection, yield estimation, temporal generalization, and spectral-domain adaptation in real-world orchard settings.

Table 1 contrasts CampanetaOrangeFruit with other publicly available citrus datasets, emphasizing its unique combination of drone-based pixel-aligned multispectral imagery, multi-temporal coverage, and instance-level annotations of individual oranges. While existing collections such as CitDet, YOLO-tiny, YOLOv8-Scm, and SootyMold rely primarily on ground-level RGB imagery captured under single-date or controlled conditions, CampanetaOrangeFruit provides five-band (RGB + R, G, RE, NIR) UAV imagery from three separate flight dates. This configuration enables cross-spectral analysis, temporal generalization, and robust benchmarking of fruit-detection and yield-estimation models in real orchard environments.

3. Data Description

3.1. Study area

Fig. 1 shows the geographic extent of the citrus orchard in Corbera, Valencia, Spain, where multispectral flights were conducted.

Table 1
Comparison of datasets.

Dataset Name	Year	Crop Type	Spectral Bands	Num. of images	Num. of annotations	Labeling Scheme	Target Classes	Notes
Campaneta OrangeFruit (this work)	2025	Citrus orchard (orange)	RGB + R, G, RE, NIR	2 750	301 232	YOLOv5	Orange	UAV based, pixel-aligned, cross spectral imagery
CitDet	2023	Citrus (HLB-affected orchards)	RGB	579	Over 32 000	COCO / YOLO	Orange	Ground-based RGB imagery (on-tree and fallen citrus fruits) for fruit detection benchmarking
YOLOC-tiny Citrus Dataset	2024	Citrus (navel orange, Ehime Jelly orange, Harumi tangerine)	RGB	3 025	10 653	YOLOv7-tiny	Citrus at multiple ripeness levels	Ground-based RGB imagery of multiripeness citrus fruits
YOLOv8-Scm Citrus Dataset	2025	Citrus (sunburn detection)	RGB	2 099	5 390	YOLOv8	Healthy vs. sunburned oranges	Ground-based RGB imagery for classification of healthy vs. sunburned citrus fruits
SootyMold Citrus Canopy Dataset	2023	Citrus canopy	RGB (night-vision)	1 000	4 500	YOLOv5 / YOLOv7	Healthy vs sooty-mold-infected leaves	Ground-based RGB dataset for detecting sooty mold on citrus canopies

The region is characterized by a Mediterranean climate, with mild winters and hot, dry summers, and rainfall concentrated mainly in autumn. Reported long-term averages for the nearby coastal Valencia area indicate an annual mean temperature of ~17–19 °C and annual precipitation of ~430–455 mm, depending on the reference station/dataset.

3.2. Repository structure

In Fig. 2, we present the complete directory structure of the Campaneta-Orange-Fruit dataset. The root directory contains five main subfolders – RGB/, G/, R/, RE/, and NIR/ – corresponding to the spectral bands captured during the UAV flight. Each folder includes the image files (.JPG for RGB and .TIF for multispectral bands), the associated YOLO-format annotation files (.txt), and a classes.txt file defining the object categories [19].

Additionally, the figures/ folder includes two illustrative images – *Figure_1.png* showing the study area and UAV flight coverage in Corbera (Valencia), and *Figure_2.png* summarizing the directory structure of annotated instances across spectral bands – while the scripts/ directory provides two Python utilities (*estimate_band_homographies.py* and *transfer_yolo_labels.py*) for computing inter-band homographies and transferring YOLO annotations, ensuring accurate pixel-level alignment across all multispectral modalities.

3.3. Class distribution

Table 2 summarises the number of annotated instances and statistics for class Orange.



Fig. 1. Study area map illustrating the orchard boundary and UAV flight coverage in Corbera, Valencia (latitude 39.1769 N, longitude 0.3538 W).

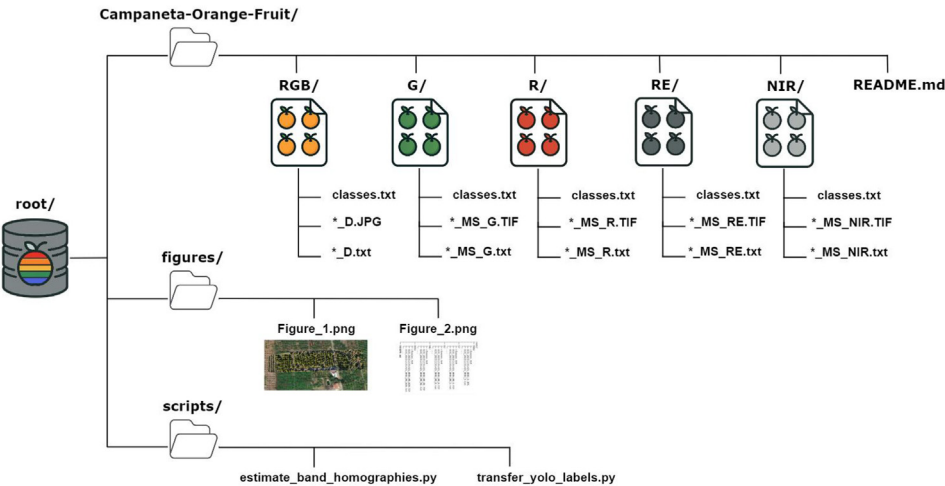


Fig. 2. Repository structure of directories and files of the dataset.

Table 2
Distribution of annotated objects per spectral band.

Band	Class name	Instances (labels)	Number of images	Mean labels per image	Minimum labels per image	Maximum labels per image
RGB	Orange	80,446	550	146.27	1	776
G	Orange	55,384	550	100.70	1	714
NIR	Orange	54,912	550	99.84	1	714
RE	Orange	55,158	550	100.29	1	713
R	Orange	55,332	550	100.60	1	714

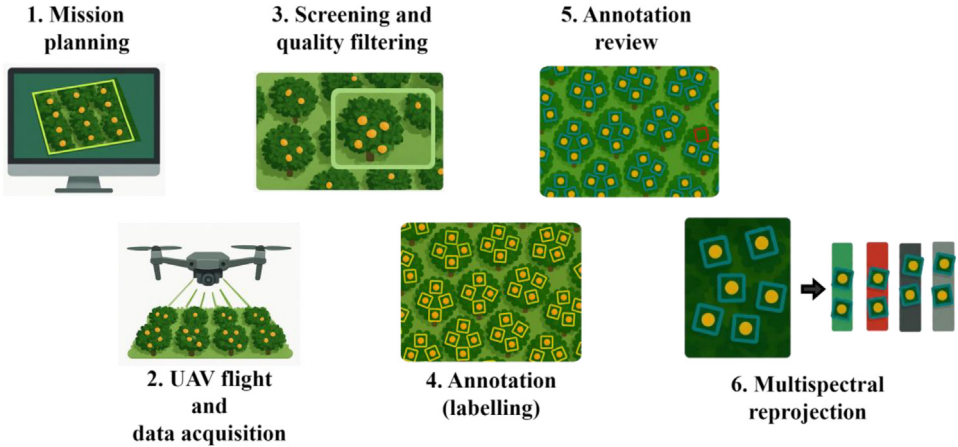


Fig. 3. Dataset construction workflow followed, from mission planning to multispectral label reprojection.

4. Experimental Design, Materials and Methods

4.1. Dataset construction workflow

The data acquisition process began with the definition of the target area, which was incorporated into a pre-planned UAV mission to ensure systematic coverage of the orange orchard. The DJI Mavic 3 Multispectral was configured to capture both RGB and multispectral imagery across four narrow bands — Green (G), Red (R), Red-Edge (RE), and Near-Infrared (NIR) — providing complementary spectral information for analysis.

Following the UAV flight, all collected images were screened for quality and content, retaining only those where oranges were clearly visible and properly exposed. The selected subset was then annotated using Labellmg, an open-source labeling tool.

The annotation team performed a detailed review of each RGB image, zooming in to accurately delineate visible oranges. These annotations were then transferred to the multispectral bands using the provided reprojection scripts, ensuring geometric consistency across all modalities.

Fig. 3 illustrates the structured workflow applied during the data acquisition and labeling stages, from mission planning to expert annotation review, culminating in the reprojection of annotations onto the multispectral bands to ensure cross-spectral consistency.

4.1.1. UAV flight and data acquisition

The image acquisition campaign was conducted under stable and cloudless conditions at mid-day to minimize shadows and illumination variability. According to this, the DJI Mavic 3 Multispectral mission was executed using a 2D single-grid flight plan designed for nadir imaging and uniform orchard coverage. A double-grid configuration was not required, as the primary objective was high-resolution fruit detection rather than 3D canopy modeling. The camera was oriented in a nadir position (gimbal angle -90°) throughout the mission to minimize geometric distortion and ensure consistent top-down perspective across all captures. The UAV flew at a constant altitude of 14 m above ground level with 80% frontal and 70% lateral overlap, at a ground speed of approximately 1.2 m/s. The total flight time for the mapped area was approximately 40 min (including takeoff and landing), resulting in 550 synchronized RGB and multispectral captures under stable midday illumination conditions.

The onboard imaging system integrates five synchronized 4/3" CMOS sensors: one 20-MP RGB camera and four monochromatic 5-MP sensors centred at 560 nm (G), 650 nm (R), 730 nm

(RE), and 860 nm (NIR), each with a 10-nm full-width half-maximum (FWHM). The resulting ground sampling distance (GSD) was approximately 0.65 cm/pixel, allowing individual fruit detection from UAV height.

A total of 550 captures were selected from the mission, each containing one RGB frame and four multispectral bands synchronized in time and geometry.

4.2. Radiometric calibration

The DJI Mavic 3 Multispectral system provides radiometrically calibrated imagery through its integrated sunlight sensor and factory calibration parameters. During acquisition, the on-board upward-facing irradiance sensor continuously measured incident solar radiation, enabling per-capture compensation for illumination variability. The multispectral bands (G, R, RE, NIR) were stored as 16-bit TIFF images containing radiometrically corrected reflectance values proportional to surface reflectance. Standard manufacturer-provided calibration metadata (including gain, exposure time, and irradiance normalization factors embedded in the image headers) were preserved without additional post-processing. No empirical reflectance panel calibration was applied, as flights were conducted under stable, cloudless midday conditions to minimize illumination changes. Therefore, the dataset provides radiometrically consistent band measurements suitable for vegetation index computation, cross-spectral modeling, and reflectance-based feature extraction, while users requiring absolute reflectance validation across campaigns may optionally perform additional panel-based normalization.

4.3. Annotation methodology

Fruit labeling was performed using LabelImg, an open-source tool compatible with the YOLO annotation format. The labeling procedure followed a three-stage verification protocol to ensure consistency and accuracy:

1. Four trained annotators independently labeled each RGB image, identifying every visible orange.
2. Two additional reviewers then cross-checked the annotations, resolving discrepancies and generating the final consensus version.
3. Bounding boxes were saved in YOLOv5 format, producing one *.txt file per image.

Each annotation file contains one line per object with five normalized values:

<class> <x_center> <y_center> <width> <height>

Where:

- <class_id> = 0 (Orange)
- <x_center>, <y_center> = normalized bounding box center coordinates in the range [0, 1]
- <width>, <height> = normalized bounding box width and height in the range [0, 1]

The RGB annotations served as the reference geometry for the multispectral bands. Using the provided Python scripts, bounding boxes were later reprojected onto the G, R, RE, and NIR images via the homographies estimated between sensors, ensuring pixel-level consistency across all spectral modalities.

As seen in Fig. 4, the dataset includes UAV images with varying densities of annotated oranges, reflecting the natural variability of fruit distribution across the orchard. Panels (a–c) illustrate canopies ranging from sparsely to densely labeled regions, which demonstrate differences in tree structure, canopy coverage, and fruit clustering captured during the UAV flights.

As shown in Fig. 5, the dataset contains orange instances with noticeable variation in size, illumination, and occlusion. Panels (a–c) display examples where fruits appear under different lighting conditions, degrees of shadowing, and levels of partial obstruction by leaves or branches, highlighting the dataset's visual diversity and its suitability for testing model robustness.

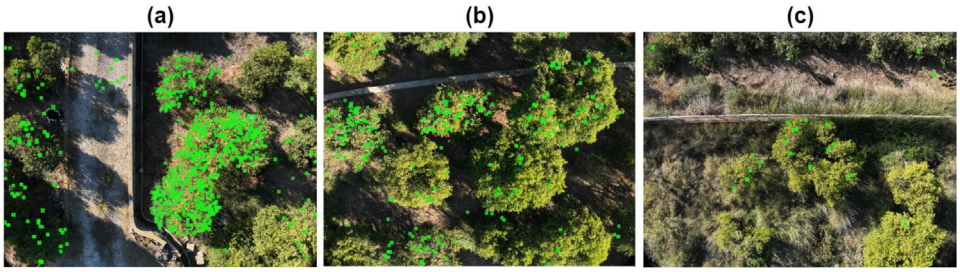


Fig. 4. Pictures with different density of labels.

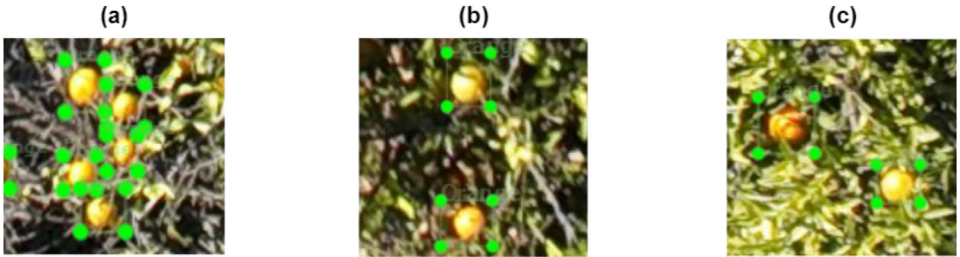


Fig. 5. Pictures with different examples of instances.

4.4. Label reprojection across spectral bands

To ensure geometric consistency across the RGB and multispectral layers, a homography-based reprojection process was implemented.

Homography estimation. For each RGB-band pair, keypoint correspondences were detected using the ORB algorithm, followed by RANSAC filtering to remove outliers. The resulting 3×3 homography matrix captures the geometric relationship between the RGB camera and each multispectral sensor, accounting for lens distortion, field-of-view differences, and gimbal offsets.

To ensure full reproducibility of the homography estimation process, we report the exact configuration used in all experiments. For each RGB-to-band pair (G, R, RE, NIR), images were first converted to grayscale and processed at a downscale factor of 0.5. Keypoint correspondences were extracted with ORB using a feature limit of 4000 keypoints per image. Binary descriptors were matched with a brute-force matcher using Hamming distance and cross-check enabled. Homographies were then estimated with `cv2.findHomography` using the RANSAC method and a reprojection threshold of 3.0 pixels. A minimum of 30 raw matches was required before running RANSAC; pairs below this threshold were discarded. The same parameterization was applied consistently across all RGB $\rightarrow$ (G, R, RE, NIR) transfers.

Bounding box reprojection. Each YOLO annotation $\langle X_a, Y_a, w, h \rangle$ from the RGB image was converted into pixel-space corner coordinates. These corner points were then warped using the corresponding homography matrix to reproject the bounding box onto the target spectral band. The resulting quadrilateral was clipped to the band's image extent, and boxes that fell partially or completely outside the frame, lost a substantial portion of their visible area, or became smaller than a minimum pixel threshold were excluded.

Therefore, after warping the RGB bounding-box corners via the estimated homography matrices, explicit filtering criteria were applied to guarantee geometric validity and practical detectability. A reprojected box was discarded if any of the following conditions were met: (i) the warped box became degenerate (zero or negative area); (ii) the warped box lay completely outside the target image after boundary clipping; (iii) $<25\%$ of the warped box area remained visible inside the target image; or (iv) the clipped axis-aligned bounding box had a width or

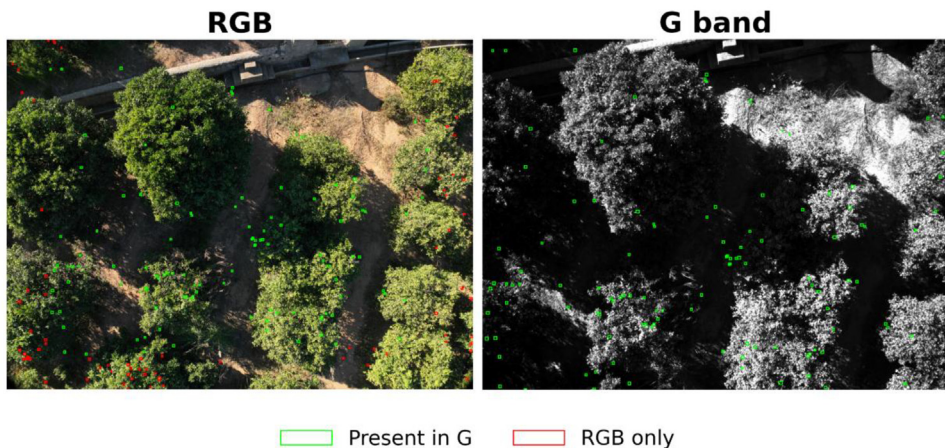


Fig. 6. RGB panel (left) shows all annotated oranges, with red boxes marking detections that fail the RGB G homography transfer; the right panel overlays the successfully reprojected boxes on the G-band radiance image, confirming where each detection remains valid.

height smaller than 2 pixels. These thresholds were applied uniformly across all spectral bands (G, R, RE, NIR) to ensure consistency and are implemented in the label-transfer procedure.

Normalization and export. Remaining boxes were renormalized with respect to the spatial dimensions of the spectral band and exported back in standard YOLO format, producing one annotation file (.txt) per multispectral image. This procedure guarantees that all spectral layers share consistent object geometry while preserving the actual parallax and optical differences between sensors.

To facilitate reuse and integration into benchmarking pipelines, a structured metadata manifest is provided in the Zenodo repository. The manifest includes, for each spectral band, image resolution, bit depth, file format, number of images, and annotation statistics. RGB images have a spatial resolution of 5280×3956 pixels (8-bit JPG), while the multispectral bands (G, R, RE, NIR) have a resolution of 2592×1944 pixels (16-bit TIFF). Each band contains 550 images, and per-band annotation counts and summary statistics are reported in Table 2. The manifest also mirrors the directory structure illustrated in Fig. 2, enabling automated dataset parsing.

Fig. 6 illustrates how the homography-driven label transfer behaves on a representative frame. The RGB image retains all original YOLO boxes, and the red overlays immediately reveal which detections disappear once warped into the G band. The adjacent G-band panel displays only the successfully mapped boxes (lime), demonstrating both the spatial consistency achieved by the transfer and the specific instances where low overlap or occlusions cause a box to be discarded. This visualization provides a qualitative sanity check that complements the quantitative band-level statistics reported earlier.

Finally, in order to promote fair and consistent benchmarking, we propose a reference evaluation protocol for fruit detection and yield-estimation research using CampanetaOrangeFruit. As the dataset comprises imagery acquired on three distinct flight dates, we recommend a date-based split as the primary experimental setup: training on two flight dates and testing on the remaining date. This configuration encourages temporal generalization and reduces spatial leakage between overlapping captures. As an alternative, a 70%/15%/15% random split at the image level may be used for within date benchmarking. For detection performance, we recommend reporting 275 mAP@0.5, mAP@0.5:0.95, Precision, Recall, and F1-score. For yield-related tasks, Mean Absolute Error (MAE) of fruit counts per image and the coefficient of determination (R^2) between predicted and ground-truth counts are suggested. These metrics align with current object-detection and precision-agriculture evaluation standards.

Limitations

The main limitations of the CampanetaOrangeFruit dataset are associated with the specific area and single UAV flight used for data acquisition. The imagery was collected over a single commercial citrus orchard located in Corbera (Valencia, Spain), which may limit the generalization of models to orchards with different canopy architectures, management practices, or environmental conditions. As the dataset represents a single acquisition campaign under specific illumination and phenological conditions, care should be taken when applying trained models to other orchards or seasons.

From a computational perspective, scaling this methodology to substantially larger orchards would increase both data volume and processing complexity. Given the native spatial resolution of the RGB images (5280×3956 pixels) and multispectral bands (2592×1944 pixels), a large-scale campaign covering several tens of hectares could easily produce several thousands of synchronized captures, resulting in tens to hundreds of gigabytes of raw imagery. This would proportionally increase storage requirements, I/O operations, and backup demands.

Photogrammetric reconstruction complexity grows superlinearly with the number of overlapping images. Structure-from-motion (SfM) alignment requires pairwise feature matching across overlapping frames, and bundle adjustment optimization scales with the number of cameras and tie points. For large datasets, memory usage and processing time may increase significantly, potentially requiring high-RAM workstations or distributed/cloud-based processing. Dense multi-view stereo reconstruction for orthomosaic generation further amplifies GPU and CPU requirements.

Although homography estimation between RGB and multispectral bands is computed per sensor pair and not globally across the orchard, processing thousands of images still increases total keypoint detection, descriptor matching, and RANSAC computations. While each operation is relatively lightweight, cumulative runtime becomes non-negligible at scale. Parallelized or batched processing would therefore be recommended for large deployments.

A further scalability constraint arises from manual annotation. Even with automated cross-spectral label transfer, the initial RGB labeling effort increases linearly with dataset size. Large-scale replication may therefore require semi-automated annotation strategies (e.g., model-assisted labeling, active learning, or pre-trained detector bootstrapping) to remain operationally feasible.

Consequently, while the proposed methodology is technically replicable in larger orchards, efficient scaling may require high-performance hardware (multi-core CPUs, ≥ 64 –128 GB RAM for large SfM projects, GPU acceleration for dense reconstruction), tiled or block-wise photogrammetric processing, and automated data-management pipelines to ensure computational tractability.

An additional limitation is that all data were acquired at a single flight 312 altitude (14 m AGL), yielding a ground sampling distance (GSD) of approximately 0.65 cm/pixel. Multi-altitude experiments were not conducted. At higher altitudes (e.g., 50 m AGL), the expected GSD would increase to approximately 1.4 cm/pixel. While mature oranges (typically 6–8 cm in diameter) would still occupy several pixels and remain theoretically detectable, reduced spatial resolution may impair the detection of smaller or partially occluded fruits and increase confusion with surrounding foliage, particularly under complex illumination conditions. Therefore, detection robustness at higher altitudes remains an open evaluation scenario and may require model retraining or architecture adjustments. Future work could systematically assess the trade-off between spatial resolution, coverage efficiency, and detection accuracy.

Ethics Statement

The work involves no human participants, no live animal experimentation, and no data collected from social media platforms. All UAV operations complied with EU Regulation 2019/947

and Spanish Royal Decree 1036/2017 on remotely piloted aircraft, including flight permissions over private agricultural land and maintenance of safe distances from inhabited areas.

CRedit Author Statement

Guillem Montalban-Faet: Conceptualization, Methodology, Data Curation, Writing - Original Draft. **Enrique Navarro-Modesto:** Data curation, Writing- review & editing. **Andoni Salcedo-Navarro:** Data curation, Writing- review & editing. **Rafael Fayos-Jordan:** Data curation, Writing- review & editing. **Pablo Benlloch-Caballero:** Data curation, Writing- review & editing. **Miguel Garcia-Pineda:** Conceptualization, Data Curation, Writing - review & editing, Supervision, Funding acquisition, Project administration. **Jaume Segura-Garcia:** Methodology, Data Curation, Writing - review & editing, Supervision, Funding acquisition, Project administration.

Declaration of AI Use

During the preparation of this work, the authors used Open AI's ChatGPT (GPT-5.0) to summarize and edit content. After using this tool, the authors reviewed and edited the content as needed and they take full responsibility for the content of the publication.

Data Availability

[Campaneta-Orange-Fruit \(Original data\)](#) (Zenodo).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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